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Welfare Implications of Public Expenditure in Indian Agriculture
New Evidence from CS-ARDL Approach

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ABSTRACT

This study explores the welfare implications of public expenditure at the subnational level. We empirically examine the efficiency of different categories of public expenditure on agriculture and irrigation using the novel cross-sectional autoregressive distributed lag (CS-ARDL) model and the data from 1981 to 2019 for 17 major agricultural states of India. The findings reveal the strong long-term positive effect in rural India of public sector expenditure on crop husbandry, agricultural research and education, soil and water conservation, irrigation, food storage and warehousing, animal husbandry and dairy development, and fisheries on total factor productivity (TFP), farm income, and poverty alleviation. Irrigation and electricity subsidies do not significantly affect outcome variables, while fertilizer subsidies showed a long-term negative impact on TFP, and credit subsidies had a positive effect on income and on the reduction of rural poverty. Other factors that were shown to have welfare implications included rainfall, literacy rate, and agricultural terms of trade (TOT), that is, the ratio of agriculture GDP to non-agriculture GDP. Policymakers should thus better target and rationalize government expenditure programs by removing unproductive input subsidies and reallocating those funds toward other types of public investment in Indian agriculture.

Keywords: Government expenditure, subsidies, welfare, Indian agriculture, CS-ARDL model

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ACRONYMS

CADF	Cross-sectional dependence Augmented Dickey-Fuller
CD	Cross-Sectional Dependence
CS-ARDL	Cross-Sectional Auto-regressive Distributive Lag
CS-DL	Cross-sectional Augmented Distributive Lag
DEAP	Data Envelopment Analysis
DPIN	Program for Decomposing Productivity Index Number
ECM	Error-correction Mechanism
GOI	Government of India
GSDP	Gross State Domestic Product
GDP	Gross Domestic Product
HAC	Heteroskedastic and Autocorrelation Consistent
LM	Lagrange Multiplier
MOSPI	Ministry of Statics and Programme Implementation
NSSO	National Sample Survey Office
RBI	Reserve Bank of India
PLFS	Periodic Labour Force Survey
TFP	Total Factor Productivity
TOT	Terms of Trade
AP	Andhra Pradesh
GUJ	Gujarat
HR	Haryana
HP	Himachal Pradesh
J&K	Jammu and Kashmir
KRT	Karnataka
KRL	Kerala
MP	Madhya Pradesh
MAH	Maharashtra
PB	Punjab
RAJ	Rajasthan
TN	Tamil Nadu
UP	Uttar Pradesh
WB	West Bengal

1 INTRODUCTION

While the world's focus has shifted back to agriculture due to concerns about feeding its nine billion inhabitants by 2050, the precise role of agriculture in economic development is still very much debated (Christiaensen, Demery, Kuhl 2011). For many rural households, improving the productivity, profitability, and sustainability of smallholder agriculture is considered to be an essential pathway out of poverty (Binswanger and Townsend 2000; de Janvry and Sadoulet 2010; Diao, Hazell, Thurlow 2010; Christiaensen, Demery, Kuhl 2011; Lipton 2012; Collier and Dercon 2014). About two-thirds of India's population resides in rural areas that are mainly dependent on agriculture, and these are areas that experience a greater concentration of poverty. Since the first Five-Year Plan, launched in 1951, the Government of India has prioritized the reduction of rural poverty. This goal has received significant emphasis in the nation's economic agenda; however, progress has been uneven both over time and geographically (Thorat and Fan 2007). In 2011/2012, an estimated 25.5 percent of rural inhabitants were found to be poor (Planning Commission of India 2013), while findings from 2018/2019 indicate that rural poverty in India has declined to 16.2 percent, which corresponds to around 143 million rural poor (Ministry of Statistics & Programme Implementation 2019).

Past research has strongly emphasized the importance of increasing India's productivity as a means of achieving agricultural growth and reducing rural poverty (Ahluwalia 1985; Srinivasan 1985; Gaiha 1989; Ghosh 1996; Fan, Hazell, Haque 2000; Thorat and Fan 2007; Fan, Gulati, Thorat 2008; Pandey and Reddy 2012; Bathla, Joshi, Kumar 2020). Inclusive market-oriented agricultural growth contributes to higher income for the farming community; this in turn creates nonfarm employment and leads to higher wages and poverty alleviation (Pandey and Reddy 2012). Since the country's independence, central and state governments have implemented several technological and institutional measures to accelerate agricultural growth (Akber and Paltasingh 2021). These have resulted in significant gains in the production and productivity of cereals, oilseeds, cotton, sugarcane, fruits, and vegetables (Chand and Parappurathu 2012). Improvement in the sector has been attributed to astute government spending, and the existing

literature highlights the role of public expenditure in growth and poverty alleviation (Ravallion and Chen 2007; Fan, Gulati, Thorat 2008; de Janvry and Sadoulet 2010; Datt, Ravallion, Murgai 2016; Bathla, Joshi, Kumar 2017; Gulati and Terway 2018; Bathla, Joshi, Kumar 2020). The 2008 study of Fan, Gulati, and Thorat confirms that public investment in research and development, irrigation, roads, and education during the Green Revolution period (1967/68 to 1977/1978)- resulted in increased productivity and poverty reduction. A study by Bathla, Joshi, and Kumar (2020) observed that the significant driver of poverty reduction in the country is public spending on irrigation, agricultural research, education, and healthcare. India has experienced a major policy debate over public expenditure in the agricultural sector, with researchers and policymakers discussing various ways of supporting agricultural output through public investment and input subsidies (Akber, Paltasingh, Mishra 2022). Public spending on farm input subsidies has always been controversial (Bathla, Joshi, Kumar 2020). The provision of input subsidies started on the recommendations of the 1964 Jha Committee (Foodgrain Prices Committee 1964). These subsidies were aimed primarily at stimulating agricultural investment, growth, and employment generation (Gulati and Sharma 1995; Gulati and Narayanan 2003); it is claimed, however, that over the years, they curtailed public investment, outlived their utility, and resulted in the overuse of natural resources (Gulati and Sharma 1995; Gulati and Bathla 2001; Fan, Hazell, Thorat 1999; Fan, Gulati, Thorat 2008; Gulati and Terway 2018; Akber 2022). This study aims to determine the welfare implications for productivity, farm income, and poverty of different categories of public expenditure. We consider four important types of input subsidies, namely those on fertilizer, irrigation, electricity, and credit and expenditures on crop production, soil conservation, food storage and warehousing, irrigation, animal husbandry and dairy development, and fisheries.

This study makes several contributions to the literature. This study analyzes the direct impact of public agricultural expenditures on productivity, farm income and poverty reduction across 17 major agricultural states in India over a 38-year period. Second, we study different categories of public expenditures in the agriculture sector in order to get a clearer picture of their effectiveness. Third, this study improves on the existing literature from a methodological point of view; for example, it checks cross-sectional dependency,

slope homogeneity, and unit roots in the data sets, which are missing from the existing literature. Therefore, there is possibility that the existing literature on the issues misses some of the characteristics of their data. Further, we used cross-sectional autoregressive distributed lag (CS-ARDL) model to check the effectiveness in long run and short run.

The present study is structured as follows. After a brief introduction, in the second section, we provide a conceptual framework of the welfare implications of public expenditure in Indian agriculture. The third section is devoted to the data and methodology used in the study, and the fourth section analyzes and discusses the empirical results. And the fifth section offers conclusion and policy implications.

2 WELFARE IMPLICATIONS OF PUBLIC EXPENDITURE IN INDIAN AGRICULTURE: A CONCEPTUAL FRAMEWORK

Public expenditure augments agricultural growth and contributes to poverty alleviation by promoting the adoption of improved technology, enhancing the availability of irrigation and other inputs, and improving wages, relative prices, and nonfarm employment (Bathla, Joshi, Kumar 2020). The increased land productivity due to spending on infrastructure, technology, and human capital can stimulate productivity growth and farm income and can thereby cause a more rapid reduction in rural poverty.

Input subsidies are an important type of public spending; they aim to incentivize farmers to use sufficient inputs to make long-term investments that stimulate the country's growth, improve farm incomes, and accelerate poverty reduction. Such subsidies, however, are a liability to the national exchequer. Our study thus aims to examine the welfare implications of different categories of public expenditure in the Indian agricultural sector. The main motive behind using the various categories of expenditure is to check the magnitude of welfare implications by different categories of public expenditure in Indian agriculture.

3 DATA AND METHODOLOGY

3.1 Data and variables

The empirical analysis of the welfare implications of public expenditure uses panel data from 1981 to 2019 for 17 major agricultural states of India. We have used three outcome variables, that is, total factor productivity (TFP), farm income, and the percentage of the rural population that is poor. Data has been compiled from various published sources, and TFP has been measured by making the input-output index. Per capita income data was collated from the Government of India's Ministry of Statistics & Programme Implementation and the *Population Projections for India and States 2011–2036* (India, Ministry of Health & Family Welfare 2020). Percentage of rural poverty has been measured by using the method put forward by the 'Tendulkar Committee, which was formed in 2009' and the data has been compiled from India's National Sample Survey Office (NSSO) and its Planning Commission. The public expenditure data on crop production, agricultural research and education, soil and water conservation, animal husbandry and dairy development, food storage and warehousing, fisheries, and irrigation are taken from the budget documents of the finance departments of the respective state governments. The study uses four major input subsidies, namely subsidies of fertilizer, irrigation, electricity, and credit. To estimate the fertilizer subsidy, the proportionate share of a state in the national consumption of fertilizers is multiplied by the total fertilizer subsidy at the all-India level. Fertilizer subsidies have been estimated using data from the Ministry of Agriculture & Farmers Welfare. Figures for irrigation subsidies were derived by subtracting receipts from operation and maintenance expenditures on irrigation (that is, revenue expenditures on major, medium, and minor irrigation) and adding 1 percent interest on capital stock and 10 percent depreciation on cumulative capital stock. Electricity subsidies were calculated by deducting the agricultural tariff from the unit cost of supply multiplied by the electricity consumption in agriculture. The data for estimating the electricity subsidy was compiled from the Central Electricity Authority of India and the Ministry of Agriculture & Farmers Welfare. Similarly, the credit subsidy was estimated by dividing the weighted average lending rates of the agricultural and non-agricultural sectors and multiplying the result by the total agricultural credit of all regional rural and commercial banks. Credit subsidy data has been taken from Reserve Bank of India

(RBI) figures. It may reflect only the interest subsidy and not the default subsidy that follows from non-payment of the interest and principal amount within the required period. Table 1A and 2A in the Appendix provides the data for these four major input subsidies at an aggregate level as well as at the state level.

The data series are converted into constant series (2011/2012 prices) using the GDP deflator. The other control variables are rainfall, agricultural terms of trade (TOT), and literacy rate (Table 1).

Table 1. Definitions, sources, and descriptive statistics of explanatory variables

Variable	Description	Source	Mean	Standard deviation
Outcome variables				
TFP	Total factor productivity (index having a range of 0, 1)	Measured using the Färe-Primont Index	0.38	0.18
Income	Per capita agricultural income (agricultural GSDP at 2011/2012 prices/total population in INR/person)	Ministry of Statistics & Programme Implementation, GoI; <i>Population Projections for India and States 2011–2036</i> ; GoI, 2019	15,127.49	5,724.52
Rural poverty	Rural poverty (%) (percentage of poor people below state-specific poverty lines, as measured by using the “Tendulkar method”)	NSSO; Planning Commission of India- various years.	34.205	17.684
Output and inputs for measuring total factor productivity (TFP)				
PR	Production of foodgrains and non-foodgrains in '000' metric tons	Agricultural Statistics at a Glance, various issues; Directorate of Economics and Statistics; Ministry of Agriculture & Farmers Welfare, GoI	13,776.5	11,337.2
Electricity consumption	Consumption of electricity for agriculture in gigawatts	State-wise Consumption of Electricity for Agriculture purposes; Agricultural Statistics at a Glance, various issues; state-wise land use statistics, Directorate of Economics and Statistics, Ministry of Agriculture & Farmers Welfare, GoI	5,493.67	6,567.55

Net irrigated area	Net irrigated area in '000 hectares	State-wise land use statistics, Directorate of Economics and Statistics, Ministry of Agriculture & Farmers Welfare, GoI	3,218.74	2,966.13
Net sown area	Net sown area in '000 hectares	State-wise land use statistics, Directorate of Economics and Statistics, Ministry of Agriculture & Farmers Welfare, GoI	7,984.69	5,864.81
Fertilizer consumption	Fertilizer consumption in '000 tons (used as an input to measure TFP)	Agricultural Statistics at a Glance, various issues, Directorate of Economics and Statistics, GoI	1,002.52	918.06
Labour	Number of persons employed in agriculture	Annual reports of Periodic Labour Force Survey (PLFS) for 2017/2018, 2018/2019, 2019/2020; NSSO reports on key indicators of employment and unemployment in India (2011/2012); Employment and Unemployment Situation in India 2004/2005; and NSS surveys/reports of previous rounds/years, Ministry of Statistics & Programme Implementation, GoI	10,951,755	7,947,141
Explanatory variables (core variables)				
Crop husbandry	Government spending on crop production	State plans of various states for various years; Ministry of Statistics & Programme Implementation, GoI	8,819.63	14,211.74
Animal husbandry and dairy development	Government spending on animal husbandry and dairy development	State plans of various states for various years; Ministry of Statistics & Programme Implementation, GoI	3,674.71	3,966.66
Research and education	Government spending on agricultural research and education	State plans of various states for various years; Ministry of Statistics & Programme Implementation, GoI	1,601.63	1,591.3
Soil and water consumption	Government spending on soil and water conservation	State plans of various states for various years; Ministry of Statistics & Programme Implementation, GoI	1,245.67	1,830.21
Food storage and warehousing	Government spending on food storage and warehousing	State plans of states for various years; Ministry of Statistics & Programme Implementation, GoI	2,819.87	5,204.49
Fisheries	Government spending on fisheries	State plans of states for various years; Ministry of Statistics & Programme Implementation, GoI	606.84	756.57
Irrigation	Government capital spending on irrigation	State plans of states for various years; Ministry of Statistics & Programme Implementation, GoI	12,797	19,469.5

Irrigation subsidy	Irrigation subsidy	State plans of states for various years; Ministry of Statistics & Programme Implementation, GoI	1,218.07	1,341.74
Fertilizer subsidy	Fertilizer subsidy	Ministry of Agriculture & Farmers Welfare	2,020	2,321.74
Power subsidy	Power subsidy	Central Electricity Authority; Ministry of Agriculture & Farmers Welfare	2,140.32	2,319.78
Credit subsidy	Credit subsidy	Reserve Bank of India (RBI)	139.37	215.91
Control variables				
TOT	Agricultural terms of trade (ratio of agricultural GDP deflator to non-agricultural GDP deflator)	Ministry of Statistics & Programme Implementation, GoI	88.6	30.43
Rainfall	Annual rainfall (millimeters)	Rainfall statistics of India, reports (various years), India Meteorological Department, Ministry of Earth Sciences, GoI	1,156.08	594.35
Literacy	Percentage of literates in the rural population (seven years old and above)	NSS Report on Household Social Consumption on Education in India: Report No 585 (75/25.2/1) and previous reports	52.83	18.621

Note: GSDP = Gross State Domestic Product; INR = Indian Rupees; GoI = Government of India; NSSO = National Sample Survey Office; We are referring to the recommendations of the 2009 Tendulkar Committee as the "Tendulkar method".

Source: Data compiled from various published sources, as mentioned in the table.

3.2. Empirical strategy

To examine the effectiveness of public expenditure, we developed three baseline models for the three different outcome variables:

$$TFP_{it} = \alpha_0 + \alpha_1 \ln PB_{it} + \alpha_2 \ln RNF_{it} + \alpha_3 \ln LR_{it} + \alpha_4 TPF_{it-1} + \varepsilon_{it} \quad (1)$$

$$\ln Y_{it} = \beta_0 + \beta_1 \ln PB_{it} + \beta_2 \ln RNF_{it} + \beta_3 \ln LR_{it} + \beta_4 \ln Y_{it-1} + \varepsilon_{it} \quad (2)$$

$$\ln RP_{it} = \pi_0 + \pi_1 \ln PB_{it} + \pi_2 \ln RNF_{it} + \pi_3 \Delta \ln TOT_{it} + \pi_4 \ln RP_{it-1} + \varepsilon_{it}, \quad (3)$$

where TFP_{it} represents total factor productivity, Y_{it} is farm income, and RP_{it} is the percentage of rural poverty. PB_{it} denotes the vector of public expenditures, RNF is the annual rainfall, LR_{it} is the literacy rate, and TOT_{it} is the agricultural terms of trade. $\ln$ stands for the natural logarithm form α, β and π are the coefficients.

3.2.1 Total factor productivity index (TFP)

Total factor productivity may be defined as the aggregate output and input index ratio.¹ Let us assume that $q_{it} = (q_{it}, \dots, q_{jit})'$ and $x_{it} = (x_{it}, \dots, x_{kit})'$ is output and input quantity vectors of firm i in period t .

TFP can thus be measured by making the ratio of total output to total index as:

$$TFP = \frac{Q_{it}}{X_{it}} \quad (4)$$

Here $Q_{it} = (q_{it})$ is an aggregate output and $X_{it} = (x_{it})$ is an aggregate input $Q(\cdot)$ and $X(\cdot)$ are the non-negative, non-decreasing, and linear homogenous aggregator functions. The associated index numbers that measure the total factor productivity of firm i in period t relative to the TFP of firm h in period s (base period) can be expressed as:

$$TFP_{hs,it} = \frac{TFP_{it}}{TFP_{hs}} = \frac{\frac{Q_{it}}{X_{it}}}{\frac{Q_{hs}}{X_{hs}}} = Q_{hs,it} / X_{hs,it} \quad (5)$$

Here, $Q_{hs,it} = Q_{it}/Q_{hs}$ is an output quantity index and $X_{hs,it} = X_{it}/X_{hs}$ is an input quantity index. TFP growth can thus be expressed by measuring the output to input growth ratio.

¹This section is drawn from O'Donnell (2011a). O'Donnell (2011b) discusses, lucidly and in detail, the methods and measurement procedures of various TFP indexes.

Different aggregator functions give rise to TFP indexes. Here, we use the Färe-Primont TFP index developed by O'Donnell (2011a, 2011b), which is the ratio of the two indexes defined by Färe & Primont (1995). In the Färe-Primont TFP index, the output index can be written as:

$$Q(q) = D_o(x_o, q_o, t_o), \quad (6)$$

and input index can be written as:

$$X(x) = D_1(x, q_o, t_o), \quad (7)$$

where x_o, q_o are the vectors of representative aggregate input and output quantities, t_o denotes the representative time period, and $D_o(.)$ and $D_1(.)$ are representative output and input distance functions. Substituting Eq. (6) and Eq. (7) into Eq. (5), TFP can be written as:

$$TFP_{hs,it} = \frac{D_o(x_o, q_{it}, t_o)}{D_o(x_o, q_{hs}, t_o)} \frac{D_1(x_{hs}, q_o, t_o)}{D_1(x_{it}, q_o, t_o)} \quad (8)$$

TFP can be estimated through the data envelopment analysis (DEA) methodology in DIPN (A Program for Decomposing Productivity Index Numbers) software.² Because DEA is based on the underpinned assumption of output and input distance functions representing production technology available in period t , which takes the form:

$$D_0(x_{it}, q_{it}, t) = \frac{(q'_{it}\alpha)}{(\gamma + x'_{it}\beta)} \quad (9)$$

² Please refer to O'Donnell (2011a: 8) for details. The estimation procedure in DPIN software is also explained in O'Donnell (2011b).

$$D_1(x_{it}, q_{it}, t) = \frac{(x'_{it}\eta)}{(q'_{it}\phi - \delta)} \quad (10)$$

The standard output-oriented DEA problem involves selecting the values of unknown parameters in Eq. (9) to minimize $OTE_{it}^{-1} = D_0(x_{it}, q_{it}, t)^{-1}$, and the input-oriented linear programming problem involves selecting the values of unknown parameters in Eq. (10) to maximize $ITE_{it} = D_1(x_{it}, q_{it}, t)^{-1}$. The resulting linear programs are derived as:

$$D_0(x_{it}, q_{it}, t)^{-1} = OTE_{it}^{-1} = \min_{\alpha, \gamma, \beta} \{ \gamma + x'_{it}\beta : \gamma l + X'\beta \geq Q'\alpha; q'_{it}\alpha = 1; \alpha \geq 0; \beta \geq 0 \} \quad (11)$$

and

$$D_1(x_{it}, q_{it}, t)^{-1} = ITE_{it} = \max_{\phi, \delta, \eta} \{ q'_{it}\phi - \delta : Q'\phi \leq \delta_t + X'\beta; x'_{it}\eta = 1; \phi \geq 0; \eta \geq 0 \}, \quad (12)$$

where Q is a $(J \times M)_t$ matrix of observed outputs, X is a $(K \times M)_t$ matrix of observed inputs and l is an $M_t \times 1$ unit vector and M_t denotes the number of observations used to estimate the frontier in period t . DPIN software used these two linear programs (LPs) to compute the TFP index.³

3.2.2 Cross-section dependency test

Panel data can be subject to cross-sectional dependency (CD), where all units in the cross-section are correlated. This CD can occur for various reasons, such as spatial/spillover or unobservable common factors in macroeconomics. To test the CD, we employed the Lagrange multiplier (LM) test given by Breusch and Pagan (1979). In the case of fixed n (number of cross-sections) and period $T \rightarrow \infty$, the Breusch and Pagan LM test can be applied to check the CD across the panels.

³Similarly, for other types of TFP indexes, the DPIN software uses a different variant of linear program formulations to estimate the various other types of TFP measurement, such as the Malmquist-hs index, Malmquist-it index, and Hicks-Moorsteen index. The details about these LPs can be found in O'Donnell (2011b).

$$LM_{BP} = T \left(\sum_{i=1}^{n-1} \sum_{j=i+1}^n \hat{\rho}_{ij}^2 \rightarrow d^2 N(N-1)/2, \right. \quad (13)$$

where $\hat{\rho}_{ij}^2$ represents the sample pairwise correlation of the residuals.

3.3.3 Panel unit root test

Pesaran's (2007) cross-sectional dependence Augmented Dickey-Fuller (CADF) test, the second-generation panel unit root test, has been applied to check the existence of unit roots in the data set. The first-generation unit root tests cannot be used here, however, because they are based on the assumption of cross-sectional independence across the cross-sections.

Pesaran (2007) examined a one-factor model with heterogeneous residual loading components; they included the average cross-section of lagged levels and first differences of the individual series to the usual Dickey-Fuller or Augmented Dickey-Fuller regressions as:

$$\Delta y_{it} = a_i + b_i y_{i,t-1} + c_i \bar{y}_{t-1} + \sum_{j=0}^P d_{ij} \Delta \bar{y}_{t-j} + \sum_{j=1}^P \delta_{ij} \Delta y_{i,t-j} + e_{it} \quad (14)$$

3.2.4 Slope homogeneity test

Although there is likely to be strong dependence across the states, each state follows different methods and designs different policies for their development; it is thus essential to test for slope heterogeneity across the states. The next step is to check the heterogeneity in the slope coefficients of Eqs. (1), (2), and (3). To that end, we applied the delta Δ_{HAC} , and the adjusted delta $\Delta_{HAC(adj.)}$ tests that were developed by Pesaran and Yamagata (2008) and improved by Blomquist and Westerlund (2013).

Initially, Swamy (1970) developed a test to check if the slope coefficients of cointegration are homogeneous. Pesaran and Yamagata (2008) later improved the test and formed two delta tests $\hat{\Delta}$, and $\hat{\Delta}_{adj}$:

$$\hat{\Delta} = \frac{\sqrt{N}(N^{-1}\tilde{S} - K)}{\sqrt{2K}} \quad (15)$$

Here K , $\tilde{S}$, and N are the number of exogenous regressors, Swamy's (1970) test statistic, and the cross-section dimension, respectively. The small sample properties of the $\hat{\Delta}$ test could be improved under the normally distributed errors on the following mean and variance-adjusted versions of $\hat{\Delta}$:

$$\hat{\Delta}_{adj} = \frac{\sqrt{N}(N^{-1}\tilde{S} - E(\tilde{z}_{iT}))}{\sqrt{var\tilde{z}_{iT}}}, \quad (16)$$

where $E(\tilde{z}_{iT}) = K$, and $var\tilde{z}_{iT} = \frac{2K(T-K-1)}{T+1}$.

These delta tests, however, do not relax the assumption of homoskedasticity and autocorrelation; Blomquist and Westerlund (2013) thus modified these delta tests and developed a heteroscedastic, and autocorrelation consistent (HAC) robust version:

$$\Delta_{HAC} = \frac{\sqrt{N}(N^{-1}\tilde{S}_{HAC} - K)}{\sqrt{2K}} \quad (17)$$

$$\Delta_{HAC(adj)} = \frac{\sqrt{N}(N^{-1}\tilde{S}_{HAC} - E(\tilde{z}_{iT}))}{\sqrt{var\tilde{z}_{iT}}} \quad (18)$$

3.2.5 Panel cointegration

For panel cointegration analysis Kao's (1999) cointegration test is employed:

$$y_{it} = \alpha_i + \beta x_{it} + \varepsilon_{it} \quad (19)$$

where α_i are the fixed effect changing across the cross-sectional observations, β are the slope parameters, y_{it} and x_{it} are independent random walks for all i . The residual series ε_{it} should be the $I(1)$ series.

3.2.6 Cross-sectional ARDL approach

We use the cross-sectional autoregressive distributed lag (CS-ARDL) approach proposed by Chudik et al. (2016) and further developed by Ditzgen (2018). This model considers unobservable common factors that may affect the states. ARDL or generalized methods of moments (GMM) approaches are commonly used in panel studies. Panel GMM techniques address endogeneity but are inefficient in the presence of cross-sectional dependence and structural breaks. The GMM system is also limited with a large N (cross-section) and large T (time dimension). Panel CS-ARDL, on the other hand, addresses the endogeneity issue, takes care of cross-sectional dependence, non-stationarity, and slope heterogeneity, and captures the long- and short-run elasticities.

The basic model for CS-DL is given below:

$$y_{it} = c_{yi} + \theta'_i X_{i,t} + \sum_{l=0}^{p_{\bar{x}}-1} \delta_{il} \Delta x_{i,t-l} + \sum_{l=0}^{p_{\bar{x}}} w_{y,il} \bar{y}_{t-l} + \sum_{l=0}^{p_{\bar{z}}} w'_{x,il} \bar{X}_{t-l} + \varepsilon_{it}, \quad (20)$$

where $\bar{X}_t = N^{-1} \sum_{i=1}^N X_{it}$, $\bar{y}_t = N^{-1} \sum_{i=1}^N y_{it}$, $p_{\bar{z}}$ is set equal to the integer part of $T^{1/3}$, denoted as $\lceil T^{1/3} \rceil$, $p = p_{\bar{x}}$, $p_{\bar{y}}$ is set to zero. The CS-DL mean group and pooled group estimator are based on Eq. (20).

The CS-ARDL estimator is based on the following regression:

$$y_{it} = c_{yi}^* + \sum_{l=1}^{p_y} \varphi_{il} y_{i,t-l} + \sum_{l=0}^{p_x} \beta'_{il} X_{i,t-l} + \sum_{l=1}^{p_z} \psi'_{il} \bar{Z}_{i,t-l} + \varepsilon_{it}^* \quad (21)$$

Where, $\bar{Z}_t = (\bar{y}_t, \bar{x}_t')', p_z = \lceil T^{1/3} \rceil$ and a choice among additional lag orders.

The CS-ARDL estimates of the individual mean-level coefficient are then given by:

$$\hat{\theta}_{CS-ARDL,i} = \frac{\sum_{l=0}^{p_x} \hat{\beta}_{il}}{1 - \sum_{l=1}^{p_y} \hat{\varphi}_{il}}, \quad (22)$$

The estimates of short-run coefficients $(\hat{\varphi}_{il}, \hat{\beta}_{il})$ are based on Eq. (21). The mean long-run effects are estimated as $N^{-1} \sum_{i=1}^N \hat{\theta}_{CS-ARDL,i}$, and the inference is based on the usual non-parametric estimator of the asymptotic variance of the mean group estimator.

4 RESULTS AND DISCUSSION

4.1 Cross-sectional dependence, panel unit root, and slope homogeneity test

Table 2 depicts the results of the cross-sectional LM test, which demonstrate a strong cross-sectional dependency across the states in the series. The null hypothesis of no correlation is rejected at a 1 percent significance level.

Table 2. Cross-sectional dependency test (Breusch and Pagan Lagrange multiplier cross-sectional dependence test)

Variable	Statistic	Prob.
TFP	838.55***	0.000
Ln Farm Income	5165.34***	0.000
LnRural poverty	2339.99***	0.000
LnCrop husbandry	3733.51***	0.000
LnAnimal husbandry and dairy development	3097.07***	0.000
LnResearch and education	3998.01***	0.000
LnSoil and water conservation	1154.39***	0.000
LnFood storage and warehousing	1167.22***	0.000
Lnfisheries	2430.92***	0.000
LnIrrigation	604.85***	0.000
LnIrrigation subsidy	1942.65***	0.000
LnFertilizer subsidy	4671.32***	0.000
LnPower Subsidy	1811.14***	0.000
LnCredit Subsidy	4968.84***	0.000
LnTOT	3113.39***	0.000
LnRainfall	457.81***	0.000
LnLiteracy	5040.48***	0.000

Note: *** indicates significance at the $p < 0.01$ level. Source: Authors' estimates

The results of the second-generation panel unit root test (Pesaran-CADF) are presented in Table 3. These results indicate that some variables are stationary at level I (0); the variables include TFP, income, public sector expenditure, fertilizer subsidy, irrigation subsidy, credit subsidy, agricultural TOT, and rainfall. Other variables, on the other hand, have unit roots and become stationary at the first difference I (1); these include rural poverty, power subsidy, and literacy rate.

Table 3. Pesaran cross-sectional augmented Dickey-Fuller panel unit root test

Variable	Level I (0)		First difference I (1)	
	T-bar	Prob.	T-bar	Prob.
TFP	-4.514***	0.000	-	-
Ln Farm Income	-2.435***	0.000	-	-
LnRural poverty	-1.116	0.990	-3.35***	0.00
LnCrop husbandry	-3.425***	0.000	-	-
LnAnimal husbandry and dairy development	-2.549***	0.000	-	-
LnResearch and education	-3.395***	0.000	-	-
LnSoil and water conservation	-2.294**	0.010	-	-
LnFood storage and warehousing	-3.439***	0.000	-	-
Lnfisheries	-3.537***	0.000	-	-
LnIrrigation	-2.835***	0.000	-	-
LnIrrigation subsidy	-2.351***	0.000	-	-
LnFertilizer subsidy	-2.788***	0.000	-	-
LnPower Subsidy	-1.439	0.927	-6.010***	0.000
LnCredit Subsidy	-3.070***	0.000	-	-
LnTOT	-2.813***	0.000	-	-
LnRainfall	-4.746***	0.000	-	-
LnLiteracy	-1.448	0.921	-2.893***	0.000

Note: ** and *** indicate statistical significance at the $p < 0.05$ and $p < 0.01$ levels.

Table 4. Slope homogeneity tests (Blomquist and Westerlund 2013)

Model	Statistic	Value	P. value
Model specification 1 (dependent variable TFP)	Δ_{HAC}	8.30	0.00
	$\Delta_{HAC(adj.)}$	10.16	0.00
Model specification 2 (dependent variable farm income)	Δ_{HAC}	12.04	0.00
	$\Delta_{HAC(adj.)}$	15.38	0.00
Model specification 3 (dependent variable rural poverty)	Δ_{HAC}	3.61	0.00
	$\Delta_{HAC(adj.)}$	4.37	0.00

Note: *** indicates significance at the $p < 0.01$ level. Source: Authors' estimates

Having confirmed the existence of cross-sectional dependency and unit roots in the data set, in the next step, we attempted to apply the slope homogeneity test. Table 4 presents the results of Blomquist and Westerlund's (2013) slope homogeneity tests. The delta HAC and Delta HAC (adjusted) results provide strong evidence of slope heterogeneity in all the model specifications. Hence, the existence of cross-

sectional dependence, mixed order of stationarity among the variables, and slope heterogeneity thus all require the cross-sectional-ARDL (CS-ARDL) approach.

4.2 Panel cointegration results

The results of Kao's (1999) panel cointegration test are shown in Table 5. They confirm the existence of long-run cointegration among the variables in all three model specifications.

Table 5. Cointegration test (Kao residual cointegration test)

Variable	Statistic	Prob.
Specification (1)	-3.900***	0.000
Specification (2)	-2.501***	0.006
Specification (3)	-1.884**	0.029

Note: ** and *** indicate statistical significance at the $p < 0.05$ and $p < 0.01$ levels. Source: Authors' estimates

4.3 Results of CS-ARDL estimation

4.3.1 Public expenditure and total factor productivity in Indian agriculture

Table 6 presents the results of short- and long-run elasticities of the CS-ARDL model for the outcome variable TFP. The short-run results demonstrate that public expenditures on crop husbandry, agricultural research and education, soil and water conservation, food storage and warehousing, and irrigation expenditure positively and significantly affect TFP productivity in Indian agriculture. The elasticity value of expenditure on crop husbandry, for example, is 0.471, suggesting that a 1 percent increase in expenditure on crop husbandry increases TFP by 0.471 percent. Similarly, the elasticity coefficient of agricultural research and education is 0.056, indicating that a 1 percent increase in expenditure on agricultural research and education boosts TFP by 0.056 percent. The short-run elasticity of soil and water conservation turns out to be 0.031. The elasticity value of food storage and warehousing (0.034) signifies that a 1 percent increase in expenditure on food storage and warehousing is projected to increase TFP by 0.034 percent. Similarly, the elasticity of irrigation expenditure is 0.030, suggesting that a 1 percent increase in irrigation expenditure increases TFP by 0.030 percent in the short run.

The empirical results also show that input subsidies in terms of irrigation, fertilizer, power, and credit subsidy do not significantly impact TFP in the short run. The elasticity values of irrigation, fertilizer, and credit subsidies are positive, but no significant impact has been observed. The estimated elasticity of other explanatory variables suggests the positive and statistically significant effect of rainfall at a 5 percent significance level.

The lagged value of the error-correction mechanism is -0.705 and is statistically significant at a 1 percent significance level. This indicates that adjustment to the long-run equilibrium process takes more than one year if any deviations occur due to shocks in the short run.

Table 6. CS-ARDL results (dependent variable TFP)

Short run			
Variable	Coeff.	Std error	Prob.
TFP (-1)	-0.935	0.629	0.141
$\Delta \text{LnCrop husbandry}$	0.471**	0.205	0.022
$\Delta \text{LnResearch and education}$	0.056***	0.020	0.000
$\Delta \text{LnSoil and water conservation}$	0.031*	0.017	0.069
$\Delta \text{LnFood storage and ware housing}$	0.034*	0.020	0.089
$\Delta \text{LnIrrigation}$	0.030**	0.013	0.027
$\Delta \text{LnIrrigation subsidy}$	0.036	0.013	0.196
$\Delta \text{LnFertilizer subsidy}$	-0.028	0.018	0.134
$\Delta \text{LnPower subsidy}$	-0.001	0.010	0.906
$\Delta \text{LnCredit subsidy}$	0.024	0.029	0.402
$\Delta \text{LnRainfall}$	0.297***	0.103	0.000
$\Delta \text{LnLiteracy}$	0.011	0.082	0.889
ECM-1	-0.705***	0.106	0.000
Long run			
Variable	Coeff.	Std error	Prob.
LnCrop husbandry	0.670**	0.301	0.026
$\text{LnResearch and education}$	0.747***	0.138	0.000
$\text{LnSoil and water conservation}$	0.299***	0.104	0.000
$\text{LnFood storage and ware housing}$	0.217*	0.019	0.071
LnIrrigation	0.703***	0.126	0.000
$\text{LnIrrigation subsidy}$	0.285	0.178	0.106
$\text{LnFertilizer subsidy}$	-0.163***	0.107	0.000
LnPower subsidy	-0.049	0.053	0.351
LnCredit subsidy	0.009	0.141	0.487
LnRainfall	0.775***	0.210	0.000
LnLiteracy	0.327***	0.121	0.000
R-squared	0.78	--	--

Note: *, **, and *** indicate statistical significance at the $p < 0.1$, $p < 0.05$, and $p < 0.01$ levels. Source: Authors' estimates

The long-run elasticities of the CS-ARDL model are presented in Table 6. They similarly confirm the positive and significant impact of different categories of public expenditure in the long run. These positive and statistically significant elasticity values suggest that public expenditure on agriculture and irrigation induces improvements in total factor productivity in Indian agriculture. The magnitude of elasticities and significance level suggests a strong positive effect in the long run only. The elasticity value of crop husbandry, for example, turns out to be 0.670; this signifies that a 1 percent increase in expenditure in crop husbandry by public authorities increases TFP by 0.670 percent, which is higher than the short-run effect. Likewise, the elasticity value of agricultural research and education came to 0.747, which is higher than the short-run estimated elasticity value. The other categories of expenditure, such as that on soil and water conservation, food storage and warehousing, and irrigation, confirm the higher positive and significant impact of public sector expenditure in the long run. Like the short-run, the long-run elasticity values of irrigation, power, and credit subsidy confirm no significant impact.

The fertilizer subsidy, on the other hand, showed a negative impact, which may be due to the overuse of subsidized fertilizers; these findings are consistent with the studies of Gulati and Banerjee (2018) and Akber, Paltasingh, and Mishra (2022). The elasticities of other explanatory variables indicate the positive and statistically significant impact of rainfall and education on agricultural TFP in the long run.

The empirical results of CS-ARDL reaffirm the strong positive effect of different categories of public expenditure on TFP in the short and long run. These results are consistent with earlier studies by Evenson, Pray, and Rosegrant (1998), by Chand, Kumar, and Kumar (2012), and by Akber and Paltasingh (2019). However, expenditure on input subsidies showed varied impacts; the elasticities of irrigation, electricity, and credit subsidies have confirmed no significance. The insignificant effect of these input subsidies can be attributed to the inequitable distribution of input subsidies across states and farming groups. And higher environmental costs in terms of over-exploitation of groundwater, distortion in cropping patterns in favour of water-intensive crops, and future agricultural production through the increased electricity subsidy (Gulati and Sharma 1995; Acharya 1998; Vyas 2001; Fan, Gulati, Thorat 2008; Prasad 2009; Birner, Gupta, Sharma 2011; Badiani, Jeesoe, Plant 2012; Badiani and Jeesoe 2013). Fertilizer subsidies showed a negative impact

in the long run. This is in line with studies by Gulati and Banerjee (2018), and Akber, Paltasingh, and Mishra (2022), who considered the imbalanced use of nutrients and the decreasing response of crops towards fertilizer usage as being among the main reasons behind the declining growth of the country's foodgrain production.

The results also confirm the positive and significant effects of other explanatory variables such as rainfall and literacy rate.

4.3.2 Public expenditure, per capita farm income, and rural poverty in India

After reaffirming the effectiveness of different categories of public expenditure on agriculture and irrigation on TFP in Indian agriculture, we attempted to check the welfare implications of public expenditure in terms of expansion of farm income and poverty reduction in Indian states.

Table 7. CS-ARDL results (dependent variable income)

Short run			
Variable	Coeff.	Std error	Prob.
LnIncome (-1)	0.841***	0.111	0.000
ΔLnCrop husbandry	0.057***	0.011	0.000
ΔLnResearch and education	0.044*	0.023	0.057
ΔLnSoil and water conservation	0.018*	0.01	0.072
ΔLnFood storage and ware housing	0.007***	0.064	0.000
ΔLnIrrigation	0.015**	0.007	0.049
ΔLn Animal husbandry and dairy development	0.026	0.03	0.378
ΔLn Fisheries	0.014	0.017	0.393
ΔLnIrrigation subsidy	0.035	0.022	0.878
ΔLnFertilizer subsidy	-0.023	0.052	0.655
ΔLnPower subsidy	0.012	0.016	0.45
ΔLnCredit subsidy	0.055*	0.032	0.091
ΔLnRainfall	0.020*	0.012	0.097
ΔLnLiteracy	0.808**	0.316	0.011
ECM-1	-0.841***	0.111	0.000
Long run			
Variable	Coeff.	Std error	Prob.
LnCrop husbandry	0.601***	0.163	0.000
LnResearch and education	0.632***	0.133	0.000
LnSoil and water conservation	0.081*	0.047	0.085
LnFood storage and ware housing	0.004*	0.002	0.051
LnIrrigation	0.606***	0.204	0.000

Ln Animal husbandry and dairy development	0.019*	0.011	0.085
Ln Fisheries	0.013***	0.004	0.000
LnIrrigation subsidy	0.003	0.105	0.805
LnFertilizer subsidy	0.002	0.026	0.914
LnPower subsidy	0.004	0.008	0.552
LnCredit subsidy	0.033*	0.019	0.086
LnRainfall	0.126	0.011	0.289
LnLiteracy	0.424**	0.174	0.015
R-squared	0.65	--	--

Note: *, **, and *** indicate statistical significance at the $p < 0.1$, $p < 0.05$, and $p < 0.01$ levels. Source: Authors' estimates

Table 7 presents the results of CS-ARDL for the outcome variable of per capita farm income. The short-run results clearly suggest that public expenditure on agriculture induces improvements in farm income in the short run. The elasticity values of public expenditure in terms of crop husbandry, agricultural research and education, soil and water conservation, food storage and warehousing, and irrigation ranges between 0.007 and 0.057. This suggests that a 1 percent increase in public expenditure in the agricultural sector increases farming income by 0.007 percent and 0.054 percent, respectively. Among input subsidies, only credit subsidy showed a positive and significant impact. The elasticity value is 0.055, indicating that a 1 percent increase in credit subsidies increases farming income by 0.055 percent. In the short run, however, no significant impact has been exhibited from the other three types of input subsidies. Rainfall and literacy rates positively and significantly affect farming income in the short run.

The elasticity value of the error-correction mechanism (ECM) is negative (-0.841), which is highly significant at a 1 percent level of significance.

The long-run elasticities of the CS-ARDL model similarly confirm the positive and significant impact of public expenditure in agriculture and irrigation. The elasticity values and the significance level suggest that expanding public expenditure improves farm income in the long run. Our results signify that public sector agricultural expenditure increases farm income by increasing productivity growth in the sector; however, while comparing the long- and short-run elasticity values, a strong positive effect is observed in the long run only. This statement of a strong positive effect on income is based on the magnitude of elasticities in the short and long run. The value of elasticities ranges between 0.004 and 0.632, suggesting that a 1 percent

increase in public sector agricultural expenditure increases the probability of increased income by 0.004 percent to 0.632 percent in the long run, which is relatively higher than the short-run elasticities. Input subsidies in terms of irrigation, fertilizer, and electricity, on the other hand, do not mark any significant impact in the long run; credit subsidy, however, showed a positive and significant impact at a 10 percent significance level. The elasticities of rainfall and literacy rate also confirm the positive effect in the long run.

Table 8. CS-ARDL results (dependent variable poverty)

Short run			
Variable	Coeff.	Std error	Prob.
LnRural poverty (-1)	-0.074	0.159	0.640
Δ LnCrop husbandry	-0.092**	0.04	0.024
Δ LnResearch and education	-0.068*	0.035	0.054
Δ LnSoil health and water conservation	-0.112**	0.052	0.033
Δ LnFood storage and warehousing	-0.007*	0.004	0.082
Δ LnIrrigation	-0.040*	0.024	0.098
Δ LnAnimal husbandry and dairy development	-0.432*	0.252	0.060
Δ LnFishries	-0.058***	0.015	0.000
Δ LnIrrigation subsidy	0.015	0.125	0.901
Δ LnFertilizer subsidy	-0.054	0.107	0.615
Δ LnPower subsidy	-0.019	0.063	0.753
Δ LnCredit subsidy	-0.066***	0.018	0.000
Δ LnRainfall	-0.070***	0.026	0.000
LnTOT	-0.025**	0.01	0.011
ECM-1	-0.745***	0.159	0.000
Long run			
Variable	Coeff.	Std error	Prob.
LnCrop husbandry	-0.301***	0.073	0.000
LnResearch and education	-0.588***	0.098	0.000
LnSoil and water conservation	-0.171***	0.062	0.000
LnFood storage and warehousing	-0.019	0.015	0.202
LnIrrigation	-0.475***	0.135	0.000
LnAnimal husbandry and dairy development	-0.425	0.525	0.418
LnFishries	-0.040*	0.023	0.081
LnIrrigation subsidy	-0.475	0.135	0.000
LnFertilizer subsidy	-0.021	0.106	0.842
LnPower subsidy	0.008	0.058	0.878

LnCredit subsidy	-0.055**	0.027	0.046
LnRainfall	-0.005**	0.002	0.011
LnTOT	-0.360***	0.05	0.000
R-squared	0.63	--	--

Note: *, **, and *** indicate statistical significance at the $p < 0.1$, $p < 0.05$, and $p < 0.01$ levels.

Table 8 presents the short- and long-run elasticities of the CS-ARDL model for the outcome variable of rural poverty. The elasticity values of public expenditure on agriculture and irrigation confirm the negative impact in the short run. The elasticity value, for instance, turns out to be between -0.007 and -0.112, which signifies that a 1 percent increase in public expenditure in Indian agriculture reduces rural poverty by 0.007 to 0.112 percent in the short run. Input subsidies in terms of irrigation, fertilizer, and power do not render any significant impact. The signs of elasticity values are negative but not significant, even at a 10 percent significance level. The credit subsidy, however, confirms the negative and significant impact. The elasticity coefficient is -0.066, which is statistically significant at a 1 percent level. The other explanatory variables suggest that rainfall and agricultural TOT play a significant role in rural poverty reduction in the short run. The value of the error-correction mechanism is -0.745 and is highly significant at a 1 percent significance level.

The results of long-run elasticities similarly suggest that public sector expenditures on crop husbandry, agricultural research and education, soil and water conservation, food storage and warehousing, animal husbandry and dairy development, fisheries, and irrigation significantly contribute to poverty alleviation in rural India. The estimated elasticities ranging between -0.040 and -0.588 indicate that a 1 percent increase in public sector agricultural expenditure alleviates rural poverty by 0.040 to 0.588 percent in the long run. These findings are consistent with earlier studies by Fan, Hazell, and Thorat (1999), Fan, Gulati, and Thorat (2008), Gulati and Terway (2018), and Bathla, Joshi, and Kumar (2020), which found public investment to be a major driver of poverty reduction in the country. Expenditures on food storage and warehousing and on animal husbandry and dairy development, however, do not show any significant impact in the long run. The magnitude of elasticities confirms more effectiveness in the long run than in the short run. In the case

of input subsidies, only credit subsidy witnessed a significant impact. Rainfall and agricultural TOT showed a negative and statistically significant impact in the long run, suggesting that rainfall and favourable TOT are essential indicators of poverty alleviation in rural India.

Further we applied Prais-Winsten regression model with panel-corrected standard error (PCSE) in the presence of cross-sectional dependency, and heteroskedasticity to check the robustness of our models, and the results mentioned in Table 3A in the appendix confirm our models' robustness.

From the above analysis, we can draw various conclusions; first, public sector expenditure on crop husbandry, agricultural research and education, soil and water conservation, food storage and warehousing, animal husbandry and dairy development, and fisheries enhances total factor productivity and improved farm income and thus alleviates poverty in rural India. Our findings are consistent with earlier studies by Evenson, Pray, and Rosegrant (1998), Fan, Hazell, and Thorat (1999: 200), Chand, Kumar, and Kumar (2012), Gulati and Terway (2018), Akber and Paltasingh (2019), and Bathla, Joshi, and Kumar (2020). These studies found public investment to be a major contributor to productivity growth and a significant driver of poverty reduction in India. Second, the welfare implications are found more in the long run than in the short run. Third, a comparison of public expenditure elasticities shows that expenditure on agricultural research and education, followed by expenditure on irrigation and crop husbandry, demonstrates greater welfare implications than do expenditures on soil and water conservation, food storage and warehousing, animal husbandry and dairy development, and fisheries. Fourth, public expenditures on subsidizing inputs such as irrigation and power do not significantly affect productivity, income, or poverty reduction. Fertilizer subsidy negatively affects the TFP in the long run. Credit subsidy, however, positively affects farm income and helps reduce poverty. Fifth, besides public expenditures on Indian agriculture, other relevant factors such as rainfall, literacy rate, and agricultural TOT have welfare implications and can significantly increase TFP and farm income and alleviate poverty.

5 CONCLUSIONS AND POLICY IMPLICATIONS

This study analyses the welfare implications of public expenditures on Indian agriculture. It does so by examining the direct impact of public expenditure on different farming activities on total factor productivity, farm income, and rural poverty in 17 of India's major agricultural states. Using the CS-ARDL model, our study confirmed that public expenditure on agriculture and irrigation significantly contributes to the economy. The other significant determinants are rainfall, literacy rate, and favourable agricultural TOT.

Our empirical results demonstrate that public sector expenditures on crop husbandry, agricultural research and education, soil and water conservation, food storage and warehousing, animal husbandry and dairy development, irrigation, and fisheries improve TFP, farm income, and poverty alleviation in rural India in the short and long run. More welfare implications, however, have been observed in only the long run. A comparison of public expenditure elasticities signifies that expenditure on agricultural research and education, followed by expenditure on irrigation and crop husbandry, demonstrate stronger welfare implications than do expenditures on soil and water conservation, food storage and warehousing, animal husbandry and dairy development, and fisheries. In the case of input subsidies, only credit subsidies effectively induced improvements in farm income and poverty reduction. In terms of policy implications, therefore, there is a need to call for better-targeted policies and the rationalizing of these input subsidies by weighing their welfare effects against their cost to the exchequer. The need of the hour, on the other hand, is a large-scale resource infusion into public expenditures, particularly into the expansion of agricultural research and education, surface irrigation, crop husbandry, and soil and water conservation.

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APPENDIX

Table 1A Agricultural Input Subsidies in India (1981-2019)

Year	Fertilizer subsidy	Agricultural power subsidy	Credit interest subsidy	Irrigation subsidy
1981	137355	405733	19749	129865
1982	141948	385598	19456	131880
1983	146715	368564	19169	133951
1984	151661	354206	18887	136081
1985	156794	342162	18611	138271
1986	162121	332132	18340	140522
1987	167651	323859	18074	142836
1988	173391	317128	17813	145215
1989	179350	311759	17557	147661
1990	185537	307599	17306	150176
1991	191961	304519	17060	152761
1992	198632	302412	16819	163944
1993	144538	301187	10178	162774
1994	153962	300770	8014	178735
1995	165547	301097	5562	180412
1996	172837	302116	11348	188524
1997	212844	303784	10351	200659
1998	231561	310646	8152	211993
1999	255516	279313	2052	212057
2000	259160	247987	440	219611
2001	221576	295263	2538	199458
2002	186313	282570	2718	205004
2003	188825	310709	665	164004
2004	248305	278000	4117	185905
2005	275805	262289	2699	211776
2006	371240	288224	9723	237332
2007	570357	325871	9775	270804
2008	1216902	387375	20583	269741
2009	727972	423335	20639	285825
2010	689104	470586	25857	295353
2011	680024	516448	19955	282000
2012	627646	558021	34431	293354
2013	596026	447874	111215	294550
2014	604482	466640	93526	277150
2015	568613	452457	94895	272169
2016	504459	493337	73367	273429
2017	487797	482451	71733	260137
2018	496371	549112	39091	259194
2019	541749	497208	11606	270715

Note: The subsidy values are in million Rs at 2011-12 prices.

Source: Data has been calculated by compiling the data from Ministry of Agriculture & Farmers Welfare, Central Electricity Authority of India, Reserve Bank of India (RBI) and Finance Accounts (Ministry of Finance).

Table 2A. Input subsidies in Indian agriculture at state level (1980-2019)

Input Subsidy	Year	AP	Assam	Bihar	GUJ	HR	HP	J&K	KRT	KRL	MP	MAH	Odisha	PB	RAJ	TN	UP	WB
Fertilizer	1981	19793	214	5555	6027	7000	401	867	10880	3650	7224	10094	2364	15110	5127	11116	24262	7671
	1982	20350	233	5819	6316	7253	412	879	11211	3639	7568	10568	2465	15493	5373	11267	25161	7942
	1983	20922	253	6095	6619	7516	424	891	11551	3628	7928	11065	2570	15887	5631	11419	26093	8223
	1984	21510	274	6384	6937	7788	436	904	11902	3617	8306	11584	2680	16290	5901	11573	27059	8513
	1985	22115	298	6687	7270	8071	449	916	12264	3606	8702	12129	2794	16704	6185	11729	28062	8814
	1986	22736	324	7005	7619	8363	462	929	12636	3595	9116	12698	2913	17128	6482	11888	29101	9125
	1987	23375	351	7337	7985	8666	475	942	13020	3585	9551	13295	3038	17564	6793	12048	30180	9447
	1988	24033	382	7686	8368	8980	489	955	13416	3574	10006	13919	3167	18010	7119	12211	31298	9781
	1989	24708	414	8050	8770	9306	503	968	13823	3563	10482	14573	3302	18467	7460	12376	32457	10126
	1990	25403	450	8433	9191	9643	518	982	14243	3553	10982	15257	3443	18936	7819	12543	33659	10484
	1991	26117	489	8833	9632	9993	532	995	14676	3542	11505	15974	3590	19417	8194	12713	34906	10854
	1992	26851	531	9252	10095	10355	548	1009	15121	3531	12053	16724	3743	19910	8587	12884	36200	11237
	1993	19362	486	6387	6512	7974	381	798	11501	2348	8351	12754	2795	13935	6141	8930	27427	8458
	1994	19658	581	7434	7789	8167	374	977	11850	2497	9544	14182	2778	14536	7334	9135	28746	8379
	1995	21742	720	8534	7629	8550	401	1012	13718	2540	9434	14709	2956	15051	8153	9038	31613	9748
	1996	22565	848	9464	8599	9155	469	874	11971	2305	11223	14362	3318	14726	8876	9498	33758	10825
	1997	23215	1166	11142	12080	10870	512	1309	17479	2884	15487	19442	4142	17330	11537	12208	39669	12371
	1998	28969	1282	12116	12531	11488	565	1654	19456	2533	16399	21822	4456	18851	10931	12731	41891	13884
	1999	31148	1730	13573	12309	12604	542	1184	21438	3027	16750	25476	5503	20308	12408	14849	46269	16397
	2000	34693	2416	15588	10142	13997	558	1343	24959	2772	14223	23565	5363	19421	11188	14830	48176	15926
	2001	26354	2275	9953	10587	12449	531	1195	19245	2427	9821	20392	5011	17468	11610	12313	45041	14905
	2002	20496	2319	9358	8133	10578	472	980	15774	2546	8111	18301	3724	16807	7137	8814	39236	13527
	2003	22497	2518	8189	10427	11626	566	1191	13267	2350	10976	15616	4042	17948	10618	8329	36104	12561
	2004	28659	2521	10632	13930	14039	661	1178	21880	2911	14454	22139	5190	21758	11868	13144	46561	16781
	2005	39101	3092	13672	16407	16013	729	1540	26702	3099	13480	12003	6150	24494	14552	16224	51002	17545
	2006	48828	4145	20592	22924	20383	988	1860	33488	4168	22008	16351	8033	31385	19470	21587	70153	24877
	2007	71888	6160	32137	37464	30200	1446	2391	47639	5886	33046	57235	11679	42548	28095	28726	98832	34984

	2008	157348	12253	68893	78814	59648	3038	6240	112543	13989	69233	122397	26986	85361	57911	64818	203509	73922
	2009	88818	7499	37029	47454	34623	1588	3667	71278	8202	46448	85222	15109	51076	32834	34408	117866	44851
	2010	89592	7443	34096	45045	30940	1453	3395	64067	7995	46450	84420	13501	47130	35115	31749	108715	37999
	2011	88750	7319	36647	46018	37919	1366	2657	62016	8001	50238	50238	13667	50930	36000	33587	111729	42942
	2012	70120	7163	38379	34904	34257	1238	2810	39216	7161	47695	64043	12613	51524	34204	24183	118502	39636
	2013	79200	6912	31199	41165	29157	1296	2682	41422	8124	45856	71555	12773	45177	31280	23068	95283	29876
	2014	70875	7098	31757	43086	32037	1362	2699	44237	5029	42040	71650	12707	44485	32250	24698	103172	35299
	2015	65643	5514	36085	35556	30501	1305	2734	38600	4941	40842	62948	12688	45470	34082	25297	92192	34217
	2016	60462	4654	28970	34181	28183	1208	2388	32461	3520	36465	60112	11067	39091	27001	18424	86640	29631
	2017	54599	4956	30263	37307	26805	1184	2558	29937	4390	33954	60187	11242	33786	22530	17533	87877	28688
	2018	52984	5540	32118	32228	27954	1175	2181	32817	3293	37661	56225	11725	35563	27704	21115	88284	27805
	2019	58357	5334	33903	37524	28056	1214	1485	34954	3247	44363	60462	12052	40186	31915	18760	100244	29693
power	1981	15014	729	97460	33530	3891	22	181	15332	390	83167	48432	785	38570	17934	17170	14249	18878
	1982	15723	667	81801	33575	4135	22	188	16083	397	78560	48928	785	36929	17954	17768	14744	17337
	1983	16466	611	68658	33619	4395	23	196	16872	403	74209	49429	785	35359	17974	18386	15257	15922
	1984	17244	559	57627	33664	4671	23	204	17699	410	70098	49935	785	33855	17994	19026	15788	14623
	1985	18058	512	48368	33708	4964	23	212	18567	417	66215	50447	785	32415	18015	19689	16337	13430
	1986	18912	468	40597	33753	5275	24	221	19478	424	62547	50963	785	31037	18035	20374	16906	12334
	1987	19805	428	34074	33798	5607	24	230	20433	432	59083	51485	785	29717	18055	21083	17494	11327
	1988	20741	392	28599	33843	5959	24	239	21435	439	55810	52013	785	28453	18075	21817	18102	10403
	1989	21720	359	24004	33887	6333	25	249	22486	446	52719	52545	785	27243	18095	22577	18732	9554
	1990	22747	328	20148	33932	6730	25	259	23588	454	49798	53083	785	26085	18115	23363	19384	8774
	1991	23821	301	16910	33977	7153	26	270	24745	462	47040	53627	785	24975	18136	24176	20058	8058
	1992	24946	275	14193	34022	7602	26	280	25958	470	44434	54176	785	23913	18156	25018	20756	7401
	1993	26125	252	11913	34068	8079	26	292	27231	478	41973	54731	785	22896	18176	25888	21478	6797
	1994	27359	230	9999	34113	8586	27	304	28566	486	39648	55292	785	21923	18197	26790	22225	6242
	1995	28652	211	8392	34158	9125	27	316	29967	494	37452	55858	785	20990	18217	27722	22998	5733
	1996	30005	193	7044	34203	9698	28	329	31436	502	35377	56430	785	20098	18237	28687	23798	5265
	1997	31423	177	5912	34249	10306	28	342	32978	511	33418	57008	785	19243	18258	29686	24626	4835
	1998	34190	189	4339	45468	12815	30	2701	23160	944	16816	59130	897	26305	21138	33425	25445	3652

	1999	33737	144	4584	54216	13587	39	520	19546	833	13583	36063	794	27720	21869	36403	11854	3822
	2000	35650	148	4731	38344	14106	41	565	19488	897	13562	25715	661	25807	12879	38536	12932	3924
	2001	40006	241	6826	42526	15437	59	396	32527	1461	15751	34139	463	25828	17163	43467	14038	4936
	2002	42536	339	8979	39022	14910	49	552	28132	1978	15662	16359	299	25698	22046	45839	15372	4796
	2003	57008	318	4320	43639	23426	60	508	48544	2016	11252	17014	148	23136	20640	47941	5168	5571
	2004	49558	124	6225	26773	22985	69	309	30261	1120	8486	23397	229	33986	16016	42639	15171	652
	2005	42747	41	5143	18729	23133	82	401	29987	571	10721	26739	139	23959	17152	43263	18499	982
	2006	54320	15	3258	30448	25016	47	599	36587	679	4341	16644	157	33656	16425	43690	18425	3916
	2007	54071	58	2743	31253	29675	50	737	43957	768	6299	20973	175	39939	25316	50421	17773	1662
	2008	69182	70	3362	37736	31479	60	565	48812	884	9751	29447	215	35374	43239	55700	19250	2249
	2009	66928	70	3366	36476	37717	76	646	51443	958	11925	33200	198	39362	57215	58836	22549	2371
	2010	68972	78	1548	22433	31172	90	585	64338	870	8139	45117	299	36285	89807	75125	22547	3182
	2011	72537	62	2034	26850	62229	190	473	60670	952	12030	60656	367	40614	86522	56172	30427	3665
	2012	117119	77	1209	30574	37084	186	943	69372	883	19202	53737	420	45738	78180	66719	32826	3752
	2013	78075	38	1067	30742	37210	160	809	68974	1029	15695	61681	421	764	37546	70131	42575	957
	2014	91540	50	952	34050	39462	1797	911	66303	732	17867	68142	520	970	31891	66559	43676	1220
	2015	89961	17	1060	25132	43564	46	925	78788	670	16938	60108	619	1294	27103	60925	45094	215
	2016	110927	18	1218	37372	43854	191	833	91271	815	15661	53073	776	1689	32738	53078	49583	239
	2017	88683	44	1451	29881	42845	202	775	93564	870	18858	58383	1000	3618	34273	54333	53397	274
	2018	135361	56	78	34587	40672	51	849	99553	767	15360	74777	1418	971	27824	63421	52699	666
	2019	90848	23	105	35308	41807	40	858	99172	753	9011	72064	1430	7083	18733	65217	54074	682
Credit interest	1981	2305	346	1149	1377	813	69	79	2376	752	1301	1552	586	1185	988	1802	2237	831
	1982	2288	337	1117	1349	802	69	78	2319	751	1274	1531	571	1168	985	1798	2207	813
	1983	2270	327	1086	1322	790	69	78	2263	750	1247	1511	555	1151	981	1793	2178	796
	1984	2253	318	1055	1296	780	69	78	2208	748	1221	1492	540	1134	977	1789	2149	780
	1985	2236	310	1026	1270	769	68	77	2155	747	1195	1472	526	1118	973	1784	2121	763
	1986	2220	301	997	1244	758	68	77	2103	746	1170	1453	512	1101	970	1780	2092	748
	1987	2203	293	969	1219	747	68	77	2052	744	1146	1434	498	1085	966	1775	2065	732
	1988	2186	285	942	1195	737	68	76	2003	743	1122	1415	484	1069	963	1771	2037	717
	1989	2170	277	916	1170	727	68	76	1955	741	1098	1397	471	1054	959	1766	2010	702

1990	2154	269	890	1147	717	68	76	1908	740	1075	1379	459	1038	955	1762	1984	687
1991	2137	262	865	1124	707	68	75	1862	739	1053	1361	446	1023	952	1758	1958	673
1992	2121	254	841	1101	697	67	75	1817	737	1031	1343	434	1008	948	1753	1932	659
1993	1339	157	517	488	421	51	58	1200	501	611	745	285	631	518	1128	1194	332
1994	1025	124	442	356	303	43	43	1009	407	420	600	210	509	432	896	949	245
1995	771	78	282	261	191	22	30	690	266	331	503	141	331	243	664	582	174
1996	1595	164	537	484	412	50	54	1400	514	704	1047	285	667	576	1329	1189	343
1997	1413	110	404	501	353	39	64	1283	501	711	938	284	526	582	1272	1104	264
1998	1117	66	298	408	300	33	49	1086	384	548	778	184	514	479	841	868	200
1999	268	15	63	97	78	8	8	286	98	143	201	46	140	132	196	227	47
2000	59	3	12	20	17	2	2	57	23	27	41	10	30	31	42	54	12
2001	329	19	63	141	110	13	11	331	119	155	252	56	159	193	215	297	77
2002	340	21	63	136	115	15	15	360	124	163	268	54	177	205	237	343	82
2003	83	4	14	29	29	3	3	88	34	38	69	11	41	47	67	78	27
2004	539	19	112	187	168	29	12	540	166	253	401	79	260	279	438	501	133
2005	324	17	73	134	118	18	8	321	104	160	331	53	150	182	290	315	100
2006	1178	55	314	492	444	65	54	1144	420	484	1056	188	607	664	1097	1091	370
2007	1157	55	267	743	400	60	45	1139	503	519	959	182	568	670	1045	1111	352
2008	2790	128	607	952	833	130	86	2363	950	1219	1903	421	1169	1422	2502	2508	603
2009	2910	139	594	951	817	134	87	2188	1073	1214	2000	468	1133	1335	2578	2306	712
2010	3771	187	740	1145	1063	152	94	2702	1573	1262	2209	628	1560	1716	3516	2641	895
2011	3080	153	591	1024	886	107	61	1734	1109	1036	1688	384	1197	1274	2869	2045	717
2012	5235	299	1103	1728	1394	183	153	2962	1948	1587	3382	619	2042	2233	4928	3500	1136
2013	16790	892	3530	5019	4650	593	610	10368	5770	5358	10322	2083	7377	7301	15562	11420	3570
2014	12989	820	2975	4372	3827	517	581	8241	5841	4855	8517	1923	6625	6263	12530	10068	2583
2015	12494	976	2924	4797	4084	625	562	8679	5064	4791	9182	2007	6625	6713	11794	10722	2854
2016	9621	749	2339	3816	3239	431	406	6507	3674	3700	7432	1355	5053	5355	8910	8803	1976
2017	10193	744	2174	3869	3121	399	400	6332	3992	3712	6339	1260	4798	5270	9198	7835	2097
2018	5820	378	1203	2014	1589	219	210	3061	2214	1940	4271	648	2292	2782	5019	4273	1157
2019	1717	116	380	623	471	63	66	917	700	569	929	202	669	853	1633	1330	364

Irrigation	1981	14962	858	8303	14828	5477	363	1713	9934	2384	4366	22961	3107	3817	7679	3489	21668	3956
	1982	15238	900	8181	14783	5580	389	1716	10242	2403	4509	23172	3202	3889	7812	3620	22246	3999
	1983	15519	944	8061	14739	5685	417	1719	10559	2421	4657	23385	3299	3962	7947	3755	22839	4043
	1984	15804	991	7942	14695	5791	448	1723	10886	2440	4810	23600	3400	4036	8085	3896	23448	4087
	1985	16096	1039	7825	14651	5900	480	1726	11223	2459	4968	23817	3504	4112	8225	4042	24073	4132
	1986	16392	1090	7710	14607	6011	514	1729	11571	2479	5131	24036	3611	4189	8367	4194	24715	4177
	1987	16694	1144	7596	14563	6124	552	1732	11929	2498	5299	24256	3721	4268	8512	4351	25374	4223
	1988	17002	1200	7484	14519	6239	591	1736	12298	2518	5473	24479	3835	4348	8659	4514	26051	4269
	1989	17315	1259	7374	14476	6356	634	1739	12679	2537	5653	24704	3952	4430	8809	4684	26746	4315
	1990	17634	1321	7265	14432	6476	680	1742	13072	2557	5838	24931	4072	4513	8961	4859	27459	4362
	1991	17959	1386	7158	14389	6597	729	1746	13477	2577	6030	25160	4196	4598	9116	5042	28191	4410
	1992	18346	1360	7208	16077	7161	707	2063	14748	2593	6846	26641	4251	4984	11857	5335	29252	4516
	1993	18758	1034	6286	16247	6775	596	2233	15709	3177	6678	27735	4394	4946	12698	5249	24942	5315
	1994	19502	2186	5246	16805	14088	849	2188	16973	3216	6252	29946	3886	4822	12702	5533	29480	5061
	1995	20726	943	5155	20209	6683	849	2365	16764	3415	6953	33942	3724	5169	13428	5414	29588	5086
	1996	22237	1024	5615	21969	6527	874	2637	18214	2781	6422	35439	5439	4592	12651	5728	30617	5758
	1997	25917	1116	6143	24526	6624	1057	2709	18747	2744	6722	38371	5085	4911	13805	6855	29273	6056
	1998	27610	1398	6502	29320	5899	1143	3196	18321	3093	7183	38535	5605	6043	16372	7983	26941	6848
	1999	26419	1311	7803	33696	6377	1102	2887	20667	2950	6928	35938	5309	5645	15923	8476	22407	8218
	2000	31370	2209	7512	33793	6147	1163	2988	22050	3033	6798	35354	4894	5826	15829	8966	21992	9687
	2001	32016	1664	5462	29292	7448	853	2816	9628	2221	6055	33282	4960	5813	16591	8345	24815	8197
	2002	34597	1630	6021	27208	8040	1341	2863	11617	2927	6517	33683	4477	5875	14846	9229	27362	6772
	2003	35875	2225	5197	5566	4424	1379	3001	11631	3162	6928	15674	3843	7759	17351	9439	24399	6150
	2004	38338	2445	8154	6806	5976	1354	3824	13936	3423	8604	25101	3982	5868	17826	9539	24471	6259
	2005	56071	2479	7721	7216	7435	1657	3045	15579	3868	8947	30352	4147	8617	18208	8566	21256	6614
	2006	68746	2890	6869	10274	7836	2404	3186	18197	3311	8157	29512	4619	7325	17649	9256	29896	7205
	2007	92474	3030	8219	10615	9993	3345	3514	16822	4277	9272	28410	7372	7273	17073	9280	33278	6556
	2008	75512	3422	8915	14607	9767	3195	3464	13643	4474	9935	37505	8455	7484	17280	9851	35554	6680
	2009	87494	4457	9176	14843	8309	3235	3813	14719	3646	10278	39546	8911	8516	16837	10086	33736	8224
	2010	92007	5422	13010	9770	8136	3290	3779	17317	3893	10161	36572	8716	9567	15501	9814	40913	7485

2011	87300	4807	12110	10389	5602	3081	1968	16619	3688	10522	37991	7412	9413	14455	8631	41394	6619
2012	99403	5560	8011	14108	9862	3042	2149	16544	4342	9225	32872	8071	9154	14989	7500	42399	6123
2013	92692	5174	8476	13096	10440	2990	4025	16343	3824	11147	32481	10438	11059	15747	11695	38525	6398
2014	77932	5572	7908	12649	9882	3099	2275	19392	3265	10439	28107	8679	9740	15929	11541	44511	6230
2015	50810	5386	8774	13217	12062	3046	12180	22096	3818	9635	30808	13840	9797	16380	11466	42556	6298
2016	50988	4791	7225	14314	11845	3942	2765	24867	4062	12057	32629	15602	9961	16826	11198	43875	6484
2017	28669	4847	8102	12641	11580	3943	2425	28510	4411	12365	30794	18132	8859	16424	12383	50025	6026
2018	37617	5418	8089	13155	11214	5512	126	30962	4019	9843	29591	15256	8456	13987	12277	47352	6318
2019	30494	5581	4655	13188	11317	5487	1844	32638	3232	15380	32803	10035	7887	14992	14666	60474	6043

Note: The subsidy values are in million Rs at 2011-12 prices.

Source: Data has been calculated by compiling the data from Ministry of Agriculture & Farmers Welfare, Central Electricity Authority of India, Reserve Bank of India (RBI) and Finance Accounts (Ministry of Finance)

Table 3A Results of Prais-Winsten regression model with panel-corrected standard error (PCSE) with cross-sectional dependency, and heteroskedasticity

	Model specification-1		Model specification-2		Model specification-3	
	PCSE (heteroskedasticity)	PCSE (autocorrelation panel-specific)	PCSE (heteroskedasticity)	PCSE (autocorrelation panel-specific)	PCSE (heteroskedasticity)	PCSE (autocorrelation panel-specific)
Variables	Coeff. (Std error)	Coeff. (Std error)	Coeff. (Std error)	Coeff. (Std error)	Coeff. (Std error)	Coeff. (Std error)
LnCrop husbandry	0.269*** (0.074)	0.129*** (0.007)	0.037* (0.020)	0.022** (0.009)	-0.077** (0.338)	-0.054** (0.026)
LnResearch and education	0.172*** (0.011)	0.104*** (0.012)	0.264*** (0.033)	0.160*** (0.019)	-0.389*** (0.037)	-0.203*** (0.037)
LnSoil and water conservation	0.009 0.005	0.010* (0.005)	0.040** (0.020)	0.003 (0.011)	-0.039 (0.027)	-0.011 (0.019)
LnFood storage and warehousing	0.030*** 0.002	0.020*** (0.001)	0.026*** (0.006)	0.005* (0.002)	-0.036*** (0.010)	-0.030*** (0.006)
LnIrrigation	0.141*** (0.006)	0.453*** (0.050)	0.053** (0.017)	0.013** (0.016)	-0.049* (0.026)	-0.064** (0.021)
LnAnimal husbandry and dairy development	--	--	0.022 (0.026)	0.026 (0.018)	-0.026 (0.045)	-0.075 (0.040)
LnFishries	--	--	0.055** (0.020)	0.031** (0.010)	-0.004 (0.033)	-0.049** (0.023)
LnIrrigation subsidy	-0.063*** (0.009)	-0.047*** (0.010)	0.019 (0.030)	0.032** (0.016)	0.078 (0.050)	0.005 (0.034)

LnFertilizer subsidy	-0.004 (0.007)	-0.014 (0.009)	-0.105*** (0.021)	-0.031** (0.014)	0.009 (0.033)	-0.075 (0.048)
LnPower subsidy	-0.041*** (0.003)	-0.034*** (0.005)	0.0165 (0.011)	0.006 (0.006)	0.055** (0.016)	0.009 (0.015)
LnCredit subsidy	-0.031 0.004	0.005 0.004	0.036** (0.011)	0.007 (0.007)	-0.044** (0.017)	-0.012 (0.022)
LnRainfall	0.011 0.012	0.044*** (0.008)	0.322*** (0.037)	0.011 (0.013)	-0.162*** (0.056)	-0.050* (0.030)
LnLiteracy	0.047** (0.021)	0.044* 0.027	1.010*** (0.069)	0.996*** (0.054)	--	--
LnTOT	--	--	--	--	-0.025 (0.071)	-0.191** (0.077)
Constant	0.876*** (0.095)	0.630*** (0.115)	7.491*** (0.069)	5.093*** (0.229)	5.487*** (0.516)	6.861*** (0.637)
R-squared	0.617	0.601	0.747	0.926	0.782	0.833

Note: *, **, and *** indicate statistical significance at the $p < 0.09$, $p < 0.05$ and $p < 0.01$ levels.

Source: Authors' estimates

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